
CLASP: Online learning algorithms for Convex Losses And Squared Penalties

Ricardo N. Ferreira¹ João Xavier² Cláudia Soares¹

Abstract

Addressing Constrained Online Convex Optimization (COCO), we introduce CLASP (Convex Losses And Squared Penalties), a framework that minimizes cumulative loss together with squared constraint violations. We propose two variants of CLASP, CLASP-I and CLASP-F, allowing for a joint or separate handling of the static decision set and the time-varying constraints, a decoupling flexibility that affords simpler implementations when projections onto the static decision set are easy. Our theoretical analysis departs from prior work by fully leveraging the variety of *cutter operators*, and contraction properties such as the strongly quasi-nonexpansiveness, a proof strategy not previously applied in this setting. For convex losses, both CLASP algorithms achieve regret $O(T^{\max\{\beta, 1-\beta\}})$ and cumulative squared penalty $O(T^{1-\beta})$ for any $\beta \in (0, 1)$. Most importantly, for strongly convex problems, we provide the first logarithmic guarantees on both regret and cumulative squared penalty: In the strongly convex case, both CLASP algorithms guarantee that the regret is upper bounded by $O(\log T)$ and the cumulative squared penalty is also upper bounded by $O(\log T)$.

1. Introduction

Online Learning can be seen as a game between a learner and an adversary, where, at each iteration $t \in \{1, 2, 3, \dots\}$, a learner selects an action x_t from a static decision set $\mathcal{K} \subset \mathbb{R}^d$. After this selection, the adversary can choose a loss function f_t , so that the learner incurs a loss associated with its decision, i.e., $f_t(x_t)$. The objective of the learner is to perform as well as the best fixed action in hindsight. In particular, the learner wants to minimize the difference

¹Department of Computer Science, NOVA School of Science and Technology, Caparica, Portugal ²Institute for Systems and Robotics, Instituto Superior Técnico, Lisboa, Portugal. Correspondence to: Ricardo N. Ferreira <rn.ferreira@campus.fct.unl.pt>.

between its incurred cumulative loss and the cumulative loss of the best fixed action in hindsight. This quantity is denominated by regret and is defined as

$$\text{Regret}_T = \sum_{t=1}^T f_t(x_t) - \sum_{t=1}^T f_t(x_T^*). \quad (1)$$

A guarantee that the regret grows sublinearly in T means that, asymptotically, the average loss of the learner is no worse than the optimal fixed action, $x_T^* = \arg \min_{x \in \mathcal{K}} \sum_{t=1}^T f_t(x)$.

Online Convex Optimization (OCO) is the special case of Online Learning in which the learner selects a decision from a convex set $\mathcal{K}$, and the loss functions f_t are convex. The OCO setup has been extensively studied over the years (Shalev-Shwartz et al., 2012; Hazan et al., 2016; Zinkevich, 2003; Duchi et al., 2011; Bubeck & Cesa-Bianchi, 2012; Hazan & Kale, 2012; Hazan & Minasyan, 2020; Hazan & Singh, 2021), and commonly assumes that operational constraints on the set of possible actions are time-invariant and fully characterized by the fixed set $\mathcal{K}$.

However, consider that operational constraints do vary over time, e.g., at each iteration t , in addition to the loss functions, a constraint convex function g_t is revealed. In this setting, we want the learner to minimize the regret, but also incur the smallest cumulative constraint violation (CCV). This setting is commonly known in the literature as Constrained Online Convex Optimization (COCO). In this setup, the best fixed decision in hindsight is the one that minimizes the regret but also satisfies the time-varying constraints for all iterations, i.e.,

$$x_T^* = \arg \min_{x \in \mathcal{K} \cap \mathcal{C}_T} \sum_{t=1}^T f_t(x), \quad (2)$$

where $\mathcal{C}_T = \bigcap_{t=1}^T C_t$ and $C_t = \{x : g_t(x) \leq 0\}$.

Prior work addressed soft constraint violations (Mahdavi et al., 2012; Jenatton et al., 2016; Yuan & Lamperski, 2018; Yu & Neely, 2020), permitting the learner to offset violations by trading off positive and negative deviations. In this setting, the cumulative constraint violation is defined as $\text{CCV}_T = \sum_{t=1}^T g_t(x_t)$. However, this metric retains a fundamental deficiency: negative deviations can counterbalance positive violations, therefore concealing the severity or frequency of constraint violations.

Many applications demand a clearer and monotonic measure of constraint violation that informs about the total severity of the constraint violations. For these reasons, the scientific community has shifted and concentrated on the hard metrics

$$CCV_{T,1} = \sum_{t=1}^T g_t^+(x_t), \text{ or } CCV_{T,2} = \sum_{t=1}^T (g_t^+(x_t))^2, \quad (3)$$

where $g_t^+(x) = \max\{0, g_t(x)\}$. These metrics are non-decreasing in T , therefore avoiding the deficiency intrinsic to the soft constraint violation metric CCV_T . **The focus of this paper is on the hard metric $CCV_{T,2}$.**

The squared penalty can be useful in applications where large violations of the constraints are a reason for concern. For example, in model predictive control, the inequality $g(x) \leq 0$ might be encoding a current saturation limit on an actuator, say, torque or thrust; here, small deviations are tolerable, but large deviations can become dangerous. As another example, in robot trajectory planning, the inequality might represent a signed distance to a harmful region. Here, violating the constraint means the robot is at risk, a risk that might not scale linearly in the sense that every extra unit of penetration does not cost the same and might be best accounted for by a quadratic increase. Yet another example might occur in a vehicle following a mobile target. Here, the constraint might model the wish to stay within a given radius of the target. If the distance becomes too large, the vehicle might lose sight of the target and fail the tracking mission.

Note that, under boundedness assumptions (rigorously defined in Section 3.3), in particular, that the feasible set has a maximum diameter D and the magnitude of the gradients of loss and constraint functions are bounded by L , the squared metric $CCV_{T,2}$ can be bounded by the linear counterpart $CCV_{T,1}$ up to a constant factor as

$$CCV_{T,2} \leq LD \cdot CCV_{T,1}.$$

However, this inequality is too loose. With this work, we show that a direct analysis of $CCV_{T,2}$ yields tighter bounds. Particularly, existing guarantees only imply $CCV_{T,1} \leq O(\sqrt{T} \log T)$ in both convex and strongly convex cases (Sinha & Vaze, 2024), whereas our direct analysis proves $CCV_{T,2} \leq O(\sqrt{T})$ for the convex setting and $O(\log T)$ for the strongly convex setting. At the same time, the squared metric remains informative for the linear metric, since by the Cauchy-Schwarz inequality, we have

$$CCV_{T,1} \leq \sqrt{T CCV_{T,2}}.$$

Thus, our framework also recover the best known bounds in the strongly convex setting. This result also reinforces the usefulness of analyzing the squared metric directly.

Contributions. In this work, we explore the use of cutter operators in the context of COCO and consider the squared-penalty violation metric $CCV_{T,2}$. We believe that our proposed algorithms and their theoretical analysis afford the following interesting features:

- **State-of-the-art guarantees.** For general convex losses, our proposed algorithms guarantee $\text{Regret}_T \leq O(T^{\max\{\beta, 1-\beta\}})$ and $CCV_{T,2} \leq O(T^{1-\beta})$, for any chosen $\beta \in (0, 1)$. This matches the rates reported by Yuan & Lamperski (2018), whose results were derived for static constraints only. In contrast, our results also allow for dynamic constraints. Furthermore, we demonstrate that, for strongly convex loss functions,

$$\text{Regret}_T \leq O(\log T) \text{ and } CCV_{T,2} \leq O(\log T).$$

To the best of our knowledge, this is the first result to guarantee logarithmic bounds on *both* regret and squared violations. Furthermore, by the Cauchy-Schwarz inequality, we demonstrate that CLASP is able to retrieve the best-known bound for $CCV_{T,1}$ in the strongly convex setting.

- **Unification and simplicity.** Prior COCO bounds relied on complex, specialized drift analyses. By leveraging the theory of *cutter operators* (Bauschke & Combettes, 2001; Cegielski & Censor, 2012), we provide a unified and modular proof framework that recovers and extends these bounds with significantly simpler logic.

- **Separability of static and dynamic constraints.** Leveraging the flexibility of cutter operators, we offer two variants of CLASP: CLASP-I iteratively projects onto the intersection $\mathcal{K} \cap C_t$ (see (2)), whereas CLASP-F projects only onto $\mathcal{K}$, dealing with C_t via a simpler subgradient cutter.

The power of cutter operators. The algorithmic and proof simplicity of the updates of both presented methods is a key theoretical finding. Historically, bounds for COCO were thought to require complex derivations. Our work demonstrates that these bounds can actually be obtained directly from the underlying strongly quasi non-expansiveness property of cutters. By distilling the analysis to its essential operator-theoretic contractive core, we provide a more streamlined and implementable framework than previous state-of-the-art.

2. Related Work

We now review the state-of-the-art in COCO. Some results in COCO consider the case of static constraints ($g_t = g$ for all t), but unknown *a priori* so they cannot be incorporated in the static decision set $\mathcal{K}$ (Mahdavi et al., 2012; Jenatton et al., 2016; Yu & Neely, 2020; Yi et al., 2021; Yuan & Lamperski, 2018). However, our focus is on the dynamic case, where g_t may vary adversarially. A summary of the main results can be found in Table 1.

Addressing dynamic constraints, Guo et al. (2022) devel-

Table 1. Summary of the main results in Constrained Online Convex Optimization. In the table, $\beta \in (0, 1)$, d denotes the dimension of the space and $\mathcal{V}$ is a problem-dependent value (for more details see (Vaze & Sinha, 2025)). “Conv-Opt” indicates solving a convex optimization problem at each round.

Reference	Regret	CCV _{T,1}	CCV _{T,2}	Complexity	Assumptions
(Yuan & Lamper-ski, 2018)	$O(T^{\max\{\beta, 1-\beta\}})$	$O(T^{1-\beta/2})$	$O(T^{1-\beta})$	Projection	Time-invariant constraints
(Guo et al., 2022)	$O(\sqrt{T})$	$O(T^{3/4})$	—	Conv-Opt	—
(Guo et al., 2022)	$O(\log T)$	$O(\sqrt{T \log T})$	—	Conv-Opt	Strongly convex losses
(Yi et al., 2023)	$O(T^{\max\{\beta, 1-\beta\}})$	$O(T^{1-\beta/2})$	—	Projection	—
(Yi et al., 2023)	$O(T^\beta)$	$O(T^{1-\beta/2})$	—	Projection	Strongly convex losses
(Sinha & Vaze, 2024)	$O(\sqrt{T})$	$O(\sqrt{T \log T})$	—	Projection	—
(Sinha & Vaze, 2024)	$O(\log T)$	$O(\sqrt{T \log T})$	—	Projection	Strongly convex losses
(Sinha & Vaze, 2024)	$O(\log T)$	$O(\log T)$	—	Projection	Strongly convex losses, Regret ≥ 0
(Wang et al., 2025)	$O(T^{\frac{3}{4}})$	$O(T^{\frac{3}{4}} \log T)$	—	Conv-Opt	—
(Wang et al., 2025)	$O(T^{\frac{2}{3}})$	$O(T^{\frac{5}{6}})$	—	Conv-Opt	Strongly convex losses
(Vaze & Sinha, 2025)	$O(\sqrt{T})$	$O(\min\{\mathcal{V}, \sqrt{T \log T}\})$	—	Projection	—
(Sinha & Vaze, 2025)	$O(\sqrt{dT \log T} + T^\beta)$	$O(dT^{1-\beta}(\log T)^2)$	—	$O(T^d)$	—
CLASP-I	$O(T^{\max\{\beta, 1-\beta\}})$	$O(T^{1-\beta/2})$	$O(T^{1-\beta})$	Projection	—
CLASP-I	$O(\log T)$	$O(\sqrt{T \log T})$	$O(\log T)$	Projection	Strongly convex losses
CLASP-F	$O(T^{\max\{\beta, 1-\beta\}})$	$O(T^{1-\beta/2})$	$O(T^{1-\beta})$	Projection	—
CLASP-F	$O(\log T)$	$O(\sqrt{T \log T})$	$O(\log T)$	Projection	Strongly convex losses

ops the Rectified Online Optimization (RECOO) algorithm. RECOO exploits a first-order approximation of the regularized Lagrangian of the underlying optimization problem at each iteration. For convex loss functions, RECOO achieves $\text{Regret}_T \leq O(\sqrt{T})$ and $\text{CCV}_{T,1} \leq O(T^{\frac{3}{4}})$, while for strongly convex loss functions, it improves to $\text{Regret}_T \leq O(\log T)$ and $\text{CCV}_{T,1} \leq O(\sqrt{T \log T})$. These results are extended in Yi et al. (2023) in the distributed setting, but whose findings can be transposed to the centralized setting, securing $\text{Regret}_T \leq O(T^{\max\{\beta, 1-\beta\}})$ and $\text{CCV}_{T,1} \leq O(T^{1-\beta/2})$ for any $\beta \in (0, 1)$ for convex loss functions; and $\text{Regret}_T \leq O(T^\beta)$ and $\text{CCV}_{T,1} \leq O(T^{1-\beta/2})$ for strongly convex loss functions.

A major leap happens in (Sinha & Vaze, 2024), who pioneered an elegant analysis technique called the Regret Decomposition Inequality. The technique is deployed

in a surrogate function inspired by the drift-plus-penalty framework (Neely, 2010). This innovative angle allows the authors to show that variants of the standard Ada-Grad algorithm can achieve $\text{Regret}_T \leq O(\sqrt{T})$ and, for the first time, $\text{CCV}_{T,1} \leq O(\sqrt{T \log T})$ for convex loss functions. For strongly convex functions, the results are $\text{Regret}_T \leq O(\log T)$ and $\text{CCV}_{T,1} \leq O(\sqrt{T \log T})$, with the CCV bound improved to $\text{CCV}_{T,1} \leq O(\log T)$ for the specialized setting of non-negative regrets. Building on this line, Vaze & Sinha (2025) propose the *Switch* algorithm, with the objective of exploiting the properties of special geometric instances of the constraint sets C_t . When the constraint sets form a sequence of nested convex bodies (i.e., $C_1 \supseteq C_2 \supseteq \dots \supseteq C_T$) and the constraint sets are all spheres or all cuboids with axis parallel with each other, then the bound on $\text{CCV}_{T,1}$ can drop to $O(1)$.

Prioritizing safety-critical applications, Sinha & Vaze (2025) present an algorithm that achieves $\text{Regret}_T \leq \tilde{O}(\sqrt{dT} + T^\beta)$ and $\text{CCV}_{T,1} \leq \tilde{O}(dT^{1-\beta})$, trading off regret in order to obtain tighter $\text{CCV}_{T,1}$ bounds.

3. Preliminaries: Geometry of Cutter Operators

We summarize the technical tools used in the analysis of our algorithms and state our assumptions.

3.1. Cutter operators

This class of operators is studied in Bauschke & Combettes (2001), although designated $\mathcal{T}$ -class, and further explored in the literature under different denominations (Yamada & Ogura, 2005; Censor & Segal, 2009; Maingé, 2010). The term “cutter operators” was later employed by Cegielski (2013), from which we adopt the terminology, since for each $x \notin \text{Fix } T \subseteq \mathbb{R}^d$, the cutter operator T allows us to define a hyperplane that cuts the space $\mathbb{R}^d$ into two half-spaces, one containing the point x and the other $\text{Fix } T$. Here, $\text{Fix } T := \{x \in \mathbb{R}^d : Tx = x\}$ denotes the set of fixed points of T .

Definition 3.1. Let $X \subseteq \mathbb{R}^d$. A map $T : X \rightarrow \mathbb{R}^d$ is a *cutter operator* if $\langle x - Tx, z - Tx \rangle \leq 0$, for all $x \in X$ and $z \in \text{Fix } T$, where $\langle \cdot, \cdot \rangle$ is the usual inner product.

Projector onto a nonempty, closed, convex set. Let $\|\cdot\|$ denote the Euclidean norm. If X is a non-empty closed, convex set, then the projector onto X , denoted by $P_X : \mathbb{R}^d \rightarrow \mathbb{R}^d$ and defined as

$$P_X(u) = \arg \min_{v \in X} \|u - v\|^2,$$

is a cutter operator (Bauschke & Combettes, 2001, Proposition 2.3).

Subgradient projector. Another example of a cutter operator is the subgradient projector $T_\phi : \mathbb{R}^d \rightarrow \mathbb{R}^d$ relative to a convex function $\phi : \mathbb{R}^d \rightarrow \mathbb{R}$, defined as

$$T_\phi x = \begin{cases} x - \frac{\phi(x)}{\|\nabla \phi(x)\|^2} \nabla \phi(x) & , \text{ if } \phi(x) > 0 \\ x & , \text{ if } \phi(x) \leq 0, \end{cases}$$

where $\nabla \phi(x)$ is a subgradient of ϕ at x , (Bauschke & Combettes, 2001, Proposition 2.3). This operator satisfies $\text{Fix } T_\phi = \{u \in \mathbb{R}^d : \phi(u) \leq 0\}$.

Definition 3.2. Let $X \subseteq \mathbb{R}^d$. For an operator $T : X \rightarrow \mathbb{R}^d$, assume that $\text{Fix } T \neq \emptyset$. We say that T is strongly quasi-nonexpansive if

$$\|Tx - z\|^2 \leq \|x - z\|^2 - \|Tx - x\|^2, \quad (4)$$

for all $x \in X$ and $z \in \text{Fix } T$.

It can be shown that an operator T with $\text{Fix } T \neq \emptyset$ is a cutter operator if and only if it is strongly quasi-nonexpansive (Cegielski, 2013, Lemma 1.2.5).

Composition of cutter operators. Let $S : \mathbb{R}^d \rightarrow \mathbb{R}^d$ and $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be cutter operators and denote their composition by $R = S \circ T$. If $x^* \in \text{Fix } S \cap \text{Fix } T$, then

$$\begin{aligned} \|Rx - x^*\|^2 &= \|S(Tx) - x^*\|^2 \\ &\leq \|Tx - x^*\|^2 - \|Rx - Tx\|^2 \\ &\leq \|x - x^*\|^2 - \|Tx - x\|^2 - \|Rx - Tx\|^2 \end{aligned} \quad (5)$$

$$\leq \|x - x^*\|^2 - \frac{1}{2} \|Rx - x\|^2, \quad (6)$$

where the first (resp. second) inequality follows from S (resp. T) being a cutter operator, and therefore strongly quasi-nonexpansive (see Eq. (4)), and the third inequality follows from the general inequality $\|u + v\|^2 \leq 2\|u\|^2 + 2\|v\|^2$, $u, v \in \mathbb{R}^d$, with $u = Rx - Tx$ and $v = Tx - x$.

This equivalence is central: we identify the Cutter property to be a sufficient condition for $O(\log T)$ violation. This bridges CLASP-I (convex intersection projection) and CLASP-F (efficient gradient update) under one framework.

3.2. Convex functions with modulus of convexity m

A function $\phi : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ is a convex function with modulus of convexity $m \geq 0$ if, for all $u, v \in \mathbb{R}^d$ and $0 \leq \lambda \leq 1$, there holds

$$\begin{aligned} \phi((1-\lambda)u + \lambda v) &\leq (1-\lambda)\phi(u) + \lambda\phi(v) \\ &\quad - m \frac{\lambda(1-\lambda)}{2} \|u - v\|^2. \end{aligned}$$

When $m = 0$, the function ϕ is usually just called convex; when $m > 0$, the function ϕ is termed strongly convex. The function ϕ is proper if $\text{dom } \phi = \{u \in \mathbb{R}^d : \phi(u) < +\infty\}$ is non-empty, and it is closed if, for any $\alpha \in \mathbb{R}$, the sublevel set $\{u \in \mathbb{R}^d : \phi(u) \leq \alpha\}$ is closed. For future reference, we now state the following well-known result.

Lemma 3.3. If $\phi : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ is a proper, closed, convex function with modulus of convexity $m \geq 0$, then

$$\phi(v) \geq \phi(u) + \langle \nabla \phi(u), v - u \rangle + \frac{m}{2} \|v - u\|^2, \quad (7)$$

for all $u, v \in \mathbb{R}^d$, with $\nabla \phi(u)$ denoting a sub-gradient of ϕ at u . In particular, if u is the minimizer of ϕ , then

$$\phi(v) \geq \phi(u) + \frac{m}{2} \|v - u\|^2, \quad (8)$$

for all $v \in \mathbb{R}^d$.

3.3. Assumptions

Throughout the paper, we work under the following assumptions, all of them standard in COCO setting:

Assumption 3.4. The action set $\mathcal{K}$ is a non-empty, compact convex subset of $\mathbb{R}^d$, with diameter $D = \max \{\|u - v\| : u, v \in \mathcal{K}\}$.

Assumption 3.5. The loss function $f_t : \mathbb{R}^d \rightarrow \mathbb{R}$ is a convex function with modulus of convexity $m \geq 0$ and that satisfies $\|\nabla f_t(u)\| \leq L$ for all $t \geq 1$ and $u \in \mathcal{K}$, where the symbol $\nabla f_t(u)$ denotes an element of the sub-differential of f_t at u .

Assumption 3.6. The constraint function $g_t : \mathbb{R}^d \rightarrow \mathbb{R}$ is a convex function that satisfies $\|\nabla g_t(u)\| \leq L$ for all $t \geq 1$ and $u \in \mathcal{K}$, where the symbol $\nabla g_t(u)$ denotes an element of the sub-differential of g_t at u .

Moreover, $\mathcal{K} \cap \bigcap_{t=1}^T C_t$ is non-empty for all $T \geq 1$, where $C_t = \{x \in \mathbb{R}^d : g_t(x) \leq 0\}$, so as to ensure the existence of the regret comparator x_T^* defined in (2).

Assumptions 3.5 and 3.6 imply that

$$\begin{aligned} |f_t(v) - f_t(u)| &\leq L \|v - u\|, \text{ and} \\ |g_t(v) - g_t(u)| &\leq L \|v - u\| \end{aligned} \quad (9)$$

for all $u, v \in \mathcal{K}$ and $t \geq 1$ (as a consequence, e.g., of the mean-value theorem (Hiriart-Urruty & Lemaréchal, 1993, Theorem 2.3.3)).

Assumption 3.7. The sequence (of step-sizes) $(\eta_t)_{t \geq 1}$ satisfies $0 < \eta_{t+1} \leq \eta_t$ for all $t \geq 1$. Hence, there exists $\theta > 0$ such that $\eta_t^2 \leq \theta \eta_t$ for all $t \geq 1$ (e.g., take $\theta = \eta_1$).

4. CLASP-I: High-Accuracy Projection

As explained before, a learner in the COCO setting deals with two types of constraints: static constraints fully described by the static decision set $\mathcal{K}$ and dynamic constraints g_t , which can be chosen adversarially. A straightforward approach for a learner is to deal with the two simultaneously. This is the basis for the first algorithm we present in the CLASP framework. We can construct a conceptually simple algorithm that, at each iteration (after the first), takes a gradient step with respect to the most recently observed loss function, and then projects onto the intersection of the static decision set with the most recently observed constraint function. We summarized it in Algorithm 1 and named it CLASP-Intersection (CLASP-I, for short).

Here, $\nabla f_t(x_t)$ represents a sub-gradient of f_t at x_t . The step-sizes η_t are chosen according to whether the loss functions are convex or strongly convex and are specified further ahead in Section 4.2. The projection step in CLASP-I is onto the set $\mathcal{K}_t = \mathcal{K} \cap C_t$, with $C_t = \{x \in \mathbb{R}^d : g_t(x) \leq 0\}$ (as defined in (2)). Note that the set $\mathcal{K}_t$ is non-empty, as per Assumption 3.6.

We now analyze CLASP-I, finding upper-bounds for both Regret_T (see (1) and (2)) and the squared-penalty violation

Algorithm 1 CLASP-Intersection (CLASP-I)

Require: action set $\mathcal{K}$, horizon $T \geq 1$, step-sizes η_t for $1 \leq t \leq T - 1$.

Choose $x_1 \in \mathcal{K}$ and observe f_1, g_1

Accumulate loss $f_1(x_1)$ and penalty $(g_1^+(x_1))^2$

for $t = 1, \dots, T - 1$ **do**

Choose $x_{t+1} = \mathcal{P}_{\mathcal{K}_t}(x_t - \eta_t \nabla f_t(x_t))$ and observe f_{t+1}, g_{t+1}

Accumulate loss $f_{t+1}(x_{t+1})$ and penalty $(g_{t+1}^+(x_{t+1}))^2$

end for

metric $\text{CCV}_{T,2}$ (see (3)). Our analysis is modular, one module bounding the metric $\text{CCV}_{T,2}$ (Section 4.1) and the other module bounding the regret (Section 4.2).

4.1. Bounding $\text{CCV}_{T,2}$

In this section, we present the module of our analysis that is dedicated to bounding the metric $\text{CCV}_{T,2}$. More precisely, we start by finding a generic bound on $\text{CCV}_{T,2}$ that is phrased in terms of the length of the step-sizes $\sum_{t=1}^T \eta_t$. Our bound, the end result of a progression of three lemmas, is stated precisely in Lemma 4.3. The lemmas hold under Assumptions 3.4 to 3.7, and their proofs are provided in the Appendix. Henceforth, $d_{\mathcal{K}_t}(x)$ denotes the distance of point x to its projection onto the set $\mathcal{K}_t$, i.e., $d_{\mathcal{K}_t}(x) = \|\mathcal{P}_{\mathcal{K}_t}(x) - x\|$.

Lemma 4.1. *In the COCO setting, for adversarially chosen convex loss functions f_t and convex constraint functions, g_t , consider Assumptions 3.4-3.7 hold. Let $\hat{x}_{t+1} = x_t - \eta_t \nabla f_t(x_t)$. Then for Algorithm 1, there holds*

$$\sum_{t=1}^T d_{\mathcal{K}_t}(\hat{x}_{t+1})^2 \leq O\left(\sum_{t=1}^T \eta_t\right).$$

Lemma 4.1 bounds the cumulative squared-distance of the intermediate step $\hat{x}_{t+1}$ (which can be sort of a virtual decision) to the convex set $\mathcal{K}_t$. The next Lemma 4.2 leverages this result and obtains a similar bound, but now relative to the decision x_t .

Lemma 4.2. *In the COCO setting, for adversarially chosen convex loss functions f_t and convex constraint functions, g_t , consider Assumptions 3.4-3.7 hold. Then for Algorithm 1, there holds*

$$\sum_{t=1}^T d_{\mathcal{K}_t}(x_t)^2 \leq O\left(\sum_{t=1}^T \eta_t\right).$$

Lemma 4.3. *In the COCO setting, for adversarially chosen convex loss functions f_t and convex constraint functions, g_t ,*

consider Assumptions 3.4-3.7 hold. Then for Algorithm 1, there holds

$$CCV_{T,2} \leq O\left(\sum_{t=1}^T \eta_t\right).$$

Lemma 4.3, the main result of this section, connects the growth of the $CCV_{T,2}$ metric with the growth of the step-sizes. The next section chooses these step-sizes for the convex and strongly convex setting and bounds the regret, completing our analysis.

4.2. Analysis of the algorithm

In this section, we provide the full analysis of CLASP-I for convex loss functions and strongly convex loss functions. The proofs for both theorems are provided in the Appendix

Theorem 4.4 (Convex losses). *In the COCO setting, for adversarially chosen convex loss functions f_t and convex constraint functions, g_t , consider Assumptions 3.4-3.7 hold. Let $\eta_t = 1/t^\beta$, with $\beta \in (0, 1)$. Then, Algorithm 1 achieves the following regret and $CCV_{T,2}$ bounds:*

$$\text{Regret}_T \leq O(T^{\max\{\beta, 1-\beta\}}) \quad \text{and} \quad CCV_{T,2} \leq O(T^{1-\beta}).$$

Theorem 4.5 (Strongly convex losses). *In the COCO setting, for adversarially chosen convex loss functions f_t , with modulus of convexity $m > 0$, and convex constraint functions, g_t , consider Assumptions 3.4-3.7 hold. Let $\eta_t = 1/(mt)$. Then, Algorithm 1 achieves the following regret and $CCV_{T,2}$ bounds:*

$$\text{Regret}_T \leq O(\log T) \quad \text{and} \quad CCV_{T,2} \leq O(\log T).$$

5. CLASP-F: Computationally Efficient Update

While CLASP-I yields tighter feasible set approximations at each iteration, there are some cases in which a projection onto the static decision set $\mathcal{K}$ admits a closed-form or finite-step solution, whereas the projection onto the time-varying set $\mathcal{K}_t$ used in the CLASP-I algorithm may not be a trivial procedure. These cases motivate the development of CLASP-F, which exploits the variety of cutter operators, dealing with the time-varying constraint via a subgradient projector and allowing a computationally efficient update.

At each iteration, we first compute an auxiliary variable $\bar{x}_t = P_{\mathcal{K}} \circ T_{g_t}(x_t)$, where we first apply the subgradient projector relative to the recently revealed constraint function g_t , followed by a projection onto the static decision set $\mathcal{K}$. Posteriorly, we use the new point and apply a gradient step with respect to the recently revealed loss function f_t and project again onto the static decision set. We summarized it in Algorithm 2 and named it CLASP-Fast (CLASP-F, for short).

Algorithm 2 CLASP-Fast (CLASP-F)

Require: action set $\mathcal{K}$, horizon $T \geq 1$, step-sizes η_t for $1 \leq t \leq T-1$.
 Choose $x_1 \in \mathcal{K}$ and observe f_1, g_1 .
 Accumulate loss $f_1(x_1)$ and penalty $(g_1^+(x_1))^2$.
for $t = 1, \dots, T-1$ **do**
 $\bar{x}_t = P_{\mathcal{K}} \circ T_{g_t}(x_t)$
 Choose $x_{t+1} = P_{\mathcal{K}}(\bar{x}_t - \eta_t \nabla f_t(\bar{x}_t))$ and observe f_{t+1}, g_{t+1}
 Accumulate loss $f_{t+1}(x_{t+1})$ and penalty $(g_{t+1}^+(x_{t+1}))^2$
end for

Here, $\nabla f_t(\bar{x}_t)$ represents a sub-gradient of f_t at $\bar{x}_t$. The step-sizes η_t are chosen according to whether the loss functions are convex or strongly convex and are specified further ahead in Section 5.3.

As shown in Section 3, the subgradient projector relative to the constraint function, T_{g_t} , and the projector onto the static decision set, $P_{\mathcal{K}}$, are both cutter operators. Furthermore, and as discussed in Section 1, note that the subgradient projector is rather simple to compute as we presume we are in the full information setting (with information about the value function and its subgradient at the current point). Thus, CLASP-F suits scenarios in which the projection onto $\mathcal{K}$ assumes a closed-form or finite-step solution. In these cases, the constraints of the problem can be handled efficiently, a characteristic particularly important in online learning applications.

We now analyze CLASP-F, deriving upper-bounds for both Regret_T (see (1) and (2)) and the squared-penalty violation metric $CCV_{T,2}$ (see (3)).

5.1. Bounding $CCV_{T,2}$

In this section, we demonstrate that the squared penalty violation can be bounded by the cumulative sum of the step-size sequence.

Lemma 5.1. *In the COCO setting, for adversarially chosen convex loss functions f_t , with modulus of convexity m , and convex constraint functions g_t , consider Assumptions 3.4-3.7 hold. Then, for Algorithm 2, there holds*

$$CCV_{T,2} \leq O\left(\sum_{t=1}^T \eta_t\right).$$

Proof. To analyze the cumulative sum of the squared penalties, we derive consequences of the two steps ($x_t \rightarrow \bar{x}_t$ and $\bar{x}_t \rightarrow x_{t+1}$) performed at each iteration of the algorithm.

Note that $\bar{x}_t = P_{\mathcal{K}} \circ T_{g_t}(x_t)$ is the composition of two cutter operators, sharing x_T^* as a fixed-point. Thus, inequality (5)

gives

$$\begin{aligned}\|\bar{x}_t - x_T^*\|^2 &\leq \|x_t - x_T^*\|^2 - \|x_t - T_{g_t}(x_t)\|^2 \\ &= \|x_t - x_T^*\|^2 - \frac{(g_t^+(x_t))^2}{\|\nabla g_t(x_t)\|^2} \\ &\leq \|x_t - x_T^*\|^2 - \frac{(g_t^+(x_t))^2}{L^2},\end{aligned}\quad (10)$$

where the last inequality follows from assumption 3.6. For the second step, $x_{t+1} = \mathcal{P}_{\mathcal{K}}(\bar{x}_t - \eta_t \nabla f_t(\bar{x}_t))$, we have

$$\begin{aligned}\|x_{t+1} - x_T^*\|^2 &= \|\mathcal{P}_{\mathcal{K}}(\bar{x}_t - \eta_t \nabla f_t(\bar{x}_t)) - x_T^*\|^2 \\ &\leq \|\bar{x}_t - \eta_t \nabla f_t(\bar{x}_t) - x_T^*\|^2 \\ &= \|\bar{x}_t - x_T^*\|^2 - 2\eta_t \langle \nabla f_t(\bar{x}_t), \bar{x}_t - x_T^* \rangle + \eta_t^2 \|\nabla f_t(\bar{x}_t)\|^2 \\ &\leq \|\bar{x}_t - x_T^*\|^2 + 2\eta_t L D + \eta_t^2 L^2 \\ &\leq \|\bar{x}_t - x_T^*\|^2 + (2D + \theta L)L\eta_t,\end{aligned}\quad (11)$$

where the first inequality results from the strongly quasi-nonexpansiveness property of cutter operators, see (4); the second inequality uses the Cauchy-Schwarz inequality, the fact that $\bar{x}_t, x_T^* \in \mathcal{K}$, and Assumptions 3.4 and 3.5; and the third inequality uses Assumption 3.7.

Combining (10) to (11) gives the inequality

$$(g_t^+(x_t))^2 \leq L^2 \|x_t - x_T^*\|^2 - L^2 \|x_{t+1} - x_T^*\|^2 + (2D + \theta L)L^3\eta_t,$$

which, summed from $t = 1$ to $t = T$, yields

$$CCV_{T,2} \leq O\left(\sum_{t=1}^T \eta_t\right). \quad (12)$$

□

5.2. Bounding the Regret

In this section, we derive a general regret bound for convex loss functions with modulus of convexity m . From this result, we will then be able to derive regret bounds for the concrete cases of convex and strongly convex loss functions.

Lemma 5.2. *In the COCO setting, for adversarially chosen convex loss functions f_t , with modulus of convexity m , and convex constraint functions g_t , consider Assumptions 3.4-3.7 hold and the step-size sequence $(\eta_t)_{t \geq 1}$ is chosen such that $\forall t \geq 1, (1/\eta_t - m) \geq 0$. Then, Algorithm 2 achieves the following general regret bound:*

$$\begin{aligned}\text{Regret}_T &\leq \sum_{t=1}^T \frac{1}{2\eta_t} \left(\|x_t - x_T^*\|^2 - \|x_{t+1} - x_T^*\|^2 \right) \\ &\quad - \sum_{t=1}^T \frac{m}{2} \|x_t - x_T^*\|^2 + \frac{3}{2} L^2 \sum_{t=1}^T \eta_t.\end{aligned}$$

Proof. We start by noting that x_{t+1} can be expressed as

$$x_{t+1} = \arg \min_{u \in \mathbb{R}^d} \phi_t(u),$$

where

$$\phi_t(u) = \langle \nabla f_t(\bar{x}_t), u \rangle + \frac{1}{2\eta_t} \|u - \bar{x}_t\|^2 + \delta_{\mathcal{K}}(u),$$

with $\delta_{\mathcal{K}} : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ denoting the indicator function of the convex set $\mathcal{K}$ (i.e., $\delta_{\mathcal{K}}(u) = 0$ if $u \in \mathcal{K}$, and $\delta_{\mathcal{K}}(u) = +\infty$, if $u \notin \mathcal{K}$). Thus, x_{t+1} is the global minimizer of a proper, closed, convex function ϕ_t with modulus $m = 1/\eta_t$. Thus, from (8), with $v = x_T^*$ and $u = x_{t+1}$, we have

$$\begin{aligned}\frac{1}{2\eta_t} \|x_{t+1} - x_T^*\|^2 &\leq \frac{1}{2\eta_t} \|\bar{x}_t - x_T^*\|^2 \\ &\quad - \frac{1}{2\eta_t} \|x_{t+1} - \bar{x}_t\|^2 + \langle \nabla f_t(\bar{x}_t), x_T^* - x_{t+1} \rangle.\end{aligned}\quad (13)$$

The next step is to upper-bound the last term on the right-hand side of (13):

$$\begin{aligned}\langle \nabla f_t(\bar{x}_t), x_T^* - x_{t+1} \rangle &= \langle \nabla f_t(\bar{x}_t), x_T^* - \bar{x}_t \rangle + \langle \nabla f_t(\bar{x}_t), \bar{x}_t - x_{t+1} \rangle \\ &\leq f_t(x_T^*) - f_t(\bar{x}_t) - \frac{m}{2} \|\bar{x}_t - x_T^*\|^2 \\ &\quad + \langle \nabla f_t(\bar{x}_t), \bar{x}_t - x_{t+1} \rangle,\end{aligned}$$

where the inequality follows from (7) applied to the term $\langle \nabla f_t(\bar{x}_t), x_T^* - \bar{x}_t \rangle$. Thus, inequality (13) evolves to

$$\begin{aligned}\frac{1}{2\eta_t} \|x_{t+1} - x_T^*\|^2 &\leq \frac{1}{2\eta_t} \|\bar{x}_t - x_T^*\|^2 + f_t(x_T^*) - f_t(\bar{x}_t) \\ &\quad - \frac{m}{2} \|\bar{x}_t - x_T^*\|^2 - \frac{1}{2\eta_t} \|x_{t+1} - \bar{x}_t\|^2 \\ &\quad + \langle \nabla f_t(\bar{x}_t), \bar{x}_t - x_{t+1} \rangle.\end{aligned}\quad (14)$$

Now, consider the generic inequality $-\frac{1}{2\eta} \|u\|^2 + \langle v, u \rangle \leq \frac{\eta}{2} \|v\|^2$, (valid for $\eta > 0$ and all $u, v \in \mathbb{R}^d$) and the fact that $\|\nabla f_t(\bar{x}_t)\| \leq L$ (from Assumption 3.5). Applying this to the last two terms of (14), we have

$$\begin{aligned}\frac{1}{2\eta_t} \|x_{t+1} - x_T^*\|^2 &\leq f_t(x_T^*) - f_t(\bar{x}_t) + \\ &\quad + \left(\frac{1}{2\eta_t} - \frac{m}{2} \right) \|\bar{x}_t - x_T^*\|^2 + \frac{\eta_t}{2} L^2.\end{aligned}\quad (15)$$

Furthermore, by inequality (7) and applying once more the fact that $\|\nabla f_t(\bar{x}_t)\| \leq L$ (from Assumption 3.5), we can rewrite the above inequality as

$$\begin{aligned}\frac{1}{2\eta_t} \|x_{t+1} - x_T^*\|^2 &\leq f_t(x_T^*) - f_t(x_t) + \\ &\quad + \left(\frac{1}{2\eta_t} - \frac{m}{2} \right) \|\bar{x}_t - x_T^*\|^2 + \frac{\eta_t}{2} L^2 \\ &\quad + L \|x_t - \bar{x}_t\| - \frac{m}{2} \|x_t - \bar{x}_t\|^2.\end{aligned}\quad (16)$$

Because $\bar{x}_t = P_{\mathcal{K}} \circ T_{g_t}(x_t)$ is the composition of two cutter operators sharing x_T^* as a fixed-point, inequality (6) allows us to upper-bound the term $\|\bar{x}_t - x_T^*\|^2$ as

$$\|\bar{x}_t - x_T^*\|^2 \leq \|x_t - x_T^*\|^2 - \frac{1}{2} \|\bar{x}_t - x_t\|^2 \quad (17)$$

Assuming that the step-size sequence $(\eta_t)_{t \geq 1}$ is chosen such that $\forall t \geq 1, (1/\eta_t - m) \geq 0$ and combining (16) and (17), we obtain

$$\begin{aligned} \frac{1}{2\eta_t} \|x_{t+1} - x_T^*\|^2 &\leq f_t(x_T^*) - f_t(x_t) + \\ &+ \left(\frac{1}{2\eta_t} - \frac{m}{2} \right) \|x_t - x_T^*\|^2 + \frac{\eta_t}{2} L^2 + \\ &+ L \|x_t - \bar{x}_t\| - \left(\frac{m}{4} + \frac{1}{4\eta_t} \right) \|x_t - \bar{x}_t\|^2. \end{aligned} \quad (18)$$

We proceed by upper-bounding the last two terms in (18) via the following generic inequality: let $z := \|x_t - \bar{x}_t\|$, $\eta_t > 0$ and L, m be nonnegative, then there holds

$$- \left(\frac{m}{4} + \frac{1}{4\eta_t} \right) z^2 + Lz \leq L^2 \eta_t.$$

From this result, we can rewrite the inequality (18) as

$$\begin{aligned} f_t(x_t) - f_t(x_T^*) &\leq \frac{1}{2\eta_t} \left(\|x_t - x_T^*\|^2 - \|x_{t+1} - x_T^*\|^2 \right) \\ &- \frac{m}{2} \|x_t - x_T^*\|^2 + \frac{3}{2} L^2 \eta_t. \end{aligned} \quad (19)$$

Summing up from $t = 1$ to T , we obtain

$$\begin{aligned} \text{Regret}_T &\leq \sum_{t=1}^T \frac{1}{2\eta_t} \left(\|x_t - x_T^*\|^2 - \|x_{t+1} - x_T^*\|^2 \right) \\ &- \sum_{t=1}^T \frac{m}{2} \|x_t - x_T^*\|^2 + \frac{3}{2} L^2 \sum_{t=1}^T \eta_t. \end{aligned} \quad (20)$$

□

5.3. Analysis of the algorithm

Having established general bounds on the regret and the squared-penalty violation metric, we can now proceed to a complete analysis of the bounds for convex loss functions (Theorem 5.3) and strongly convex loss functions (Theorem 5.4).

Theorem 5.3 (Convex losses). *In the COCO setting, for adversarially chosen convex loss functions f_t and convex constraint functions, g_t , consider Assumptions 3.4-3.7 hold. Let $\eta_t = 1/t^\beta$, with $\beta \in (0, 1)$. Then, Algorithm 2 achieves the following regret and $\text{CCV}_{T,2}$ bounds:*

$$\text{Regret}_T \leq O(T^{\max\{\beta, 1-\beta\}}) \quad \text{and} \quad \text{CCV}_{T,2} \leq O(T^{1-\beta}).$$

Theorem 5.4 (Strongly convex losses). *In the COCO setting, for adversarially chosen strongly convex loss functions f_t , with modulus of convexity $m > 0$, and convex constraint functions, g_t , consider Assumptions 3.4-3.7 hold. Let $\eta_t = 1/(mt)$. Then, Algorithm 2 achieves the following regret and $\text{CCV}_{T,2}$ bounds:*

$$\text{Regret}_T \leq O(\log T) \quad \text{and} \quad \text{CCV}_{T,2} \leq O(\log T).$$

The proofs for both theorems are provided in the Appendix and are immediate consequences of Lemmas 5.1 and 5.2 and the choice of step-size sequence.

Operator-theoretic mindset. The proofs of CLASP-I and CLASP-F feature the operator-theoretic mindset. In fact, as CLASP-I and CLASP-F adopt a template update rule in static offline optimization, alternating between performing a projected gradient step and repairing constraint violation (e.g., (Nedić, 2011; Singh & Necoara, 2024)), the main contribution lies not in a new update rule but in the operator-theoretic formulation and the analysis of squared violations for COCO.

6. Numerical Experiments

In this section, we present a synthetic experiment to evaluate the performance of our proposed algorithms. We compare CLASP-I and CLASP-F with the AdaGrad-based algorithm of Sinha & Vaze (2024), the RECOO algorithm of Guo et al. (2022), the Frank-Wolfe-based algorithm of Wang et al. (2025), and the Switch algorithm of Vaze & Sinha (2025). We report results for the cumulative loss and for the violation metrics $\text{CCV}_{T,1}$ and $\text{CCV}_{T,2}$. In line with prior COCO work, we measure $\text{CCV}_{T,2}$ *a posteriori* to illustrate how this theoretically motivated quantity behaves in practice.

Further synthetic and real-data experiments can be found in Appendix E.1. All results were averaged over 50 trials and are reported with 95% confidence intervals¹. The experiments were performed on a 2020 MacBook Air 13", with an 8-core Apple M1 processor and 16 GB of RAM.

Online Linear Regression. This experiment is similar to the one presented by Guo et al. (2022) for the setting of adversarial constraints with a synthetic dataset. The static decision set is given by $\mathcal{K} = \{x \in \mathbb{R}^{10} : 0 \leq x_i \leq 1, \text{ for } i = 1, \dots, 10\}$. The vector $x_1 \in \mathbb{R}^n$ in all algorithms was initialized such that $x_i \sim U(0, 1)$, for $1 \leq i \leq n$, where $U(a, b)$ stands for the uniform distribution supported on the interval (a, b) . At each iteration t , the loss function is given by $f_t(x) =$

¹The source code for the numerical experiments can be found in <https://github.com/ricardojnf/CLASP>.

$\|H_t x - y_t\|^2$, with $H_t \in \mathbb{R}^{4 \times 10}$, $x \in \mathbb{R}^{10}$ and $y_t \in \mathbb{R}^4$, where $H_t(i, j) \sim U(-1, 1)$, $1 \leq i \leq 4$, $1 \leq j \leq 10$, and $y_t(i) = H_t(i) \mathbf{1} + \epsilon_i$, such that $\mathbf{1} := (1, \dots, 1) \in \mathbb{R}^{10}$ and ϵ_i denotes the standard normal random variable. On the other hand, the constraint function is given by $g_t(x) = A_t x - b_t$, with $A_t \in \mathbb{R}^{4 \times 10}$ and $b_t \in \mathbb{R}^4$, where $A_t(i, j) \sim U(0, 2)$ and $b_t(i) \sim U(0, 1)$, for $1 \leq i \leq 4$ and $1 \leq j \leq 10$. Note that g_t is a vector-valued map taking values in $\mathbb{R}^4$, and thus corresponds to multiple constraints of the form $g_{t,m}(x) \leq 0$, for $1 \leq m \leq 4$. However, we can replace it with a single scalar function, which is the pointwise maximum of the constraints, $g_t(x) = \max\{g_{t,m}(x) : 1 \leq m \leq 4\}$.

Reduction of multiple constraint functions. For multiple constraint functions revealed at each iteration, one might want to take into account the cumulative squared constraint violation for all the functions, such as $CCV_{T,2} := \sum_{t=1}^T \|\hat{g}_t(x)\|^2$, where $\hat{g}_t(x) := (g_{t,1}(x), \dots, g_{t,m}(x))$ considering there are m different constraint functions revealed at each iteration. However, the replacement with a single scalar function as the pointwise maximum of the functions, i.e., $g_t(x) = \max\{g_{t,i}(x) : 1 \leq i \leq m\}$, is standard in COCO. Note that

$$\|\hat{g}_t^+(x)\|^2 = \sum_{i=1}^m (g_{t,i}^+(x))^2 \leq m ([g_t^+(x)])^2.$$

Hence, any upper bound on $\sum_t ([g_t(x)]_+)^2$ immediately implies a bound on the vector-valued metric up to the constant factor m . A similar argument can be made for the linear metric $CCV_{T,1}$.

Figure 1 reports cumulative loss, linear violation ($CCV_{T,1}$), and squared violation ($CCV_{T,2}$) over iterations. The AdaGrad-based method (blue) attains the lowest cumulative loss, but only by incurring much larger constraint violations. By contrast, CLASP-I (orange) controls both $CCV_{T,1}$ and $CCV_{T,2}$ at levels comparable to RECOO (red), while remaining simpler and more memory-efficient. The Frank-Wolfe-based algorithm (purple) and CLASP-F (green) achieve slightly lower cumulative loss than CLASP-I, RECOO, and Switch, at the cost of higher cumulative constraint violation. Switch (brown) achieves the smallest overall violations, but at the cost of higher per-iteration complexity.

Notably, on the squared violation metric $CCV_{T,2}$, the focus of our analysis, both CLASP algorithms are competitive with the best-performing baselines, confirming in practice that it can keep violation severity sublinear while maintaining reasonable regret. This statement becomes clearer when running the experiment for more rounds, as the cumulative squared penalty for CLASP-F becomes smaller than that of the Frank-Wolfe algorithm (see Section E.1 in the Appendix).

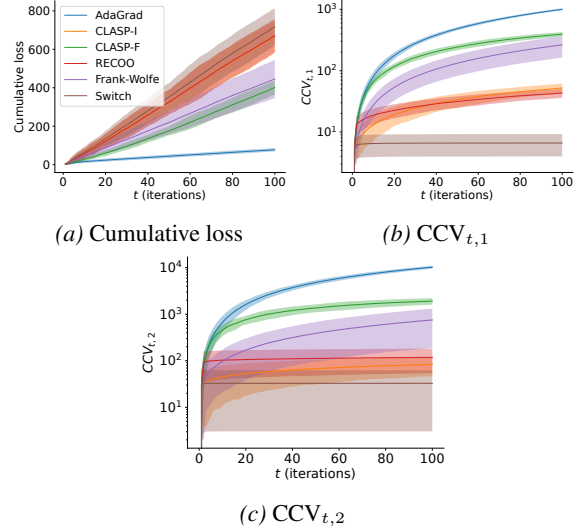


Figure 1. Online linear regression with adversarial constraints. We report (a) cumulative loss, (b) linear violation $CCV_{T,1}$, and (c) squared violation $CCV_{T,2}$. AdaGrad minimizes loss but incurs large violations. CLASP-I matches RECOO’s safety with better memory efficiency than Switch (which yields lowest violations but high complexity). Frank-Wolfe and CLASP-F incur higher violations without loss gains.

Conclusions

We present CLASP, a framework for online algorithms for Convex Losses and Squared Penalties. We focus on deriving bounds for the regret and the squared-penalty violation metric $CCV_{T,2}$, the metric of choice for safety-critical operations where large constraint violations induce disproportionate risk. Our algorithms match state-of-the-art bounds on both regret and squared-penalty violation metric $CCV_{T,2}$. In particular, for general convex loss functions, we offer a tunable trade-off between regret and $CCV_{T,2}$, matching the best performance of previous works designed for static constraints. More importantly, for strongly convex losses, we universally achieve logarithmic bounds on both regret and $CCV_{T,2}$, an advance that, to the extent of our knowledge, is established here for the first time. In this work, we exploit cutter operators, a wide class of operators used in optimization and feasibility theory. We present two algorithms, CLASP-I and CLASP-F, each of which can handle static and time-varying constraints jointly or separately. The latter being simpler to implement when the projection onto $\mathcal{K}$ assumes a closed-form or finite-step solution.

Limitations. A limitation of the CLASP framework is its reliance on squared penalties ($CCV_{T,2}$). While this aligns with stability concerns (suppressing severe outliers), it does not strictly penalize frequent, negligible violations. Thus, CLASP may not be suitable for applications requiring zero-tolerance for constraint breaches without using safety filters.

Impact Statement

Positive Impacts. By aligning online learning objectives with the quadratic stability metrics standard in control theory (LQR/MPC), CLASP facilitates the safe deployment of adaptive algorithms in physical systems—such as robotics and power grid management—where preventing severe violations is critical. Furthermore, our operator-theoretic analysis simplifies the landscape of constrained optimization, providing a modular proof framework that unifies prior results and offers a template for analyzing dynamic constraints.

Risks and mitigation. A potential risk lies in the “soft” nature of the constraint violation metrics. While it effectively penalizes severity, it technically permits small, frequent violations. In domains requiring hard safety (e.g., chemical plant pressure limits or collision avoidance), relying solely on $CCV_{T,1}$ and $CCV_{T,2}$ without an additional “hard” safety filter could be hazardous if the system cannot tolerate even minor infractions. Furthermore, as a general-purpose optimization tool, CLASP could be leveraged to improve the efficiency of harmful systems. Practitioners should assess whether their application allows for the “soft constraint” assumption inherent in quadratic penalties. For strictly critical boundaries, we recommend pairing CLASP with verified safety layers.

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A. Appendix

A.1. Proof of Lemma 4.1

Note that $x_T^* \in \mathcal{K} \cap \mathcal{C}_T \subset \mathcal{K}_t$, implying $x_T^* = P_{\mathcal{K}_t}(x_T^*)$ for $1 \leq t \leq T$. We have

$$\begin{aligned}
\|x_{t+1} - x_T^*\|^2 &= \|P_{\mathcal{K}_t}(\hat{x}_{t+1}) - P_{\mathcal{K}_t}(x_T^*)\|^2 \\
&\leq \|\hat{x}_{t+1} - x_T^*\|^2 - d_{\mathcal{K}_t}(\hat{x}_{t+1})^2 \\
&= \|x_t - \eta_t \nabla f_t(x_t) - x_T^*\|^2 - d_{\mathcal{K}_t}(\hat{x}_{t+1})^2 \\
&= \|x_t - x_T^*\|^2 - 2\eta_t \langle \nabla f_t(x_t), x_t - x_T^* \rangle + \eta_t^2 \|\nabla f_t(x_t)\|^2 - d_{\mathcal{K}_t}(\hat{x}_{t+1})^2 \\
&\leq \|x_t - x_T^*\|^2 + 2\eta_t \|\nabla f_t(x_t)\| \|x_t - x_T^*\| + \eta_t^2 \|\nabla f_t(x_t)\|^2 - d_{\mathcal{K}_t}(\hat{x}_{t+1})^2 \\
&\leq \|x_t - x_T^*\|^2 + 2LD\eta_t + L^2\eta_t^2 - d_{\mathcal{K}_t}(\hat{x}_{t+1})^2,
\end{aligned}$$

where the first inequality is due to the strongly quasi-nonexpansiveness of the operator $P_{\mathcal{K}_t}$ (see (4)), where we denote $d_{\mathcal{K}_t}(x) = \|P_{\mathcal{K}_t}(x) - x\|$; the second inequality is the Cauchy-Schwarz inequality; and the third inequality follows from Assumptions 3.4 and 3.5.

Before proceeding, we would like to indicate the pivotal role played by the strongly quasi-nonexpansiveness property of the operator $P_{\mathcal{K}_t}$. Indeed, if we used only its weaker non-expansiveness property, the right-hand side of the first inequality would read as $\|\hat{x}_{t+1} - x_T^*\|^2$, the key term $d_{\mathcal{K}_t}(\hat{x}_{t+1})^2$ no longer present. The disappearance of this term would instantly sever the reasoning of the present lemma; as the subsequent lemmas depend upon it, the entire proof would be invalidated.

Re-arranging the last inequality yields

$$\begin{aligned}
d_{\mathcal{K}_t}(\hat{x}_{t+1})^2 &\leq \|x_t - x_T^*\|^2 - \|x_{t+1} - x_T^*\|^2 + 2LD\eta_t + L^2\eta_t^2 \\
&\leq \|x_t - x_T^*\|^2 - \|x_{t+1} - x_T^*\|^2 + (2LD + \theta L^2)\eta_t,
\end{aligned}$$

where the last inequality is due to Assumption 3.7.

Summing up from $t = 1$ to T , we obtain

$$\sum_{t=1}^T d_{\mathcal{K}_t}(\hat{x}_{t+1})^2 \leq \|x_1 - x_T^*\|^2 + \sum_{t=1}^T (2LD + \theta L^2)\eta_t = O\left(\sum_{t=1}^T \eta_t\right).$$

A.2. Proof of Lemma 4.2

We have

$$\begin{aligned}
d_{\mathcal{K}_t}(x_t) &= d_{\mathcal{K}_t}(\hat{x}_{t+1} + (x_t - \hat{x}_{t+1})) \\
&\leq d_{\mathcal{K}_t}(\hat{x}_{t+1}) + \|x_t - \hat{x}_{t+1}\| \\
&= d_{\mathcal{K}_t}(\hat{x}_{t+1}) + \eta_t \|\nabla f_t(x_t)\| \\
&\leq d_{\mathcal{K}_t}(\hat{x}_{t+1}) + \eta_t L,
\end{aligned}$$

where the first inequality follows from the Lipschitz continuity of the distance function $d_{\mathcal{K}_t}$ (see (Bauschke & Combettes, 2017)), and the second inequality from Assumption 3.5.

It follows that

$$\begin{aligned}
d_{\mathcal{K}_t}(x_t)^2 &\leq 2d_{\mathcal{K}_t}(\hat{x}_{t+1})^2 + 2\eta_t^2 L^2 \\
&\leq 2d_{\mathcal{K}_t}(\hat{x}_{t+1})^2 + 2\eta_t \theta L^2,
\end{aligned}$$

where the first inequality comes from the general fact $(a + b)^2 \leq 2a^2 + 2b^2$ for $a, b \in \mathbb{R}$, and the second inequality from Assumption 3.7.

Summing up from $t = 1$ to T and using Lemma 4.1, we obtain

$$\begin{aligned} \sum_{t=1}^T d_{\mathcal{K}_t}(x_t)^2 &\leq 2 \sum_{t=1}^T d_{\mathcal{K}_t}(\hat{x}_{t+1})^2 + 2L^2\theta \sum_{t=1}^T \eta_t \\ &\leq O\left(\sum_{t=1}^T \eta_t\right). \end{aligned}$$

A.3. Proof of Lemma 4.3

Note that $|g_t^+(u) - g_t^+(v)| \leq |g_t(u) - g_t(v)|$ for all $u, v \in \mathbb{R}^n$. Hence, $|g_t^+(u) - g_t^+(v)| \leq L \|u - v\|$ for all $u, v \in \mathcal{K}$ due to Assumption 3.6 and its consequence (9).

It follows that

$$\begin{aligned} g_t^+(x_t) &\leq g_t^+(P_{\mathcal{K}_t}(x_t)) + L \|x_t - P_{\mathcal{K}_t}(x_t)\| \\ &= L d_{\mathcal{K}_t}(x_t), \end{aligned}$$

since $g_t^+(P_{\mathcal{K}_t}(x_t)) = 0$ because $P_{\mathcal{K}_t}(x_t) \in \mathcal{K}_t = \mathcal{K} \cap \{x \in \mathbb{R}^n : g_t(x) \leq 0\}$ and any point $u \in \mathcal{K}_t$ satisfies $g_t(u) \leq 0$.

Equivalently, we have $g_t^+(x_t)^2 \leq L^2 d_{\mathcal{K}_t}(x_t)^2$. Summing up from $t = 1$ to T and using Lemma 4.2, we obtain

$$\begin{aligned} \text{CCV}_{T,2} &= \sum_{t=1}^T (g_t^+(x_t))^2 \\ &\leq L^2 \sum_{t=1}^T d_{\mathcal{K}_t}(x_t)^2 \\ &\leq O\left(\sum_{t=1}^T \eta_t\right). \end{aligned}$$

B. Proof of Theorem 4.4

We start by analyzing the Regret_T . Express the projection step of CLASP-I as

$$x_{t+1} = \arg \min_{u \in \mathbb{R}^n} h_t(u),$$

where

$$h_t(u) = \langle \nabla f_t(x_t), u - x_t \rangle + \frac{1}{2\eta_t} \|u - x_t\|^2 + \delta_{\mathcal{K}_t}(u),$$

with $\delta_{\mathcal{K}_t} : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ denoting the indicator function of the convex set $\mathcal{K}_t$ (i.e., $\delta_{\mathcal{K}_t}(u) = 0$ if $u \in \mathcal{K}_t$, and $\delta_{\mathcal{K}_t}(u) = +\infty$, if $u \notin \mathcal{K}_t$).

Note that h_t is a proper, closed, and σ -strongly convex function with modulus $m = 1/\eta_t$ and global minimizer x_{t+1} . Thus, from (8), we have

$$h_t(x_T^*) \geq h_t(x_{t+1}) + \frac{1}{2\eta_t} \|x_T^* - x_{t+1}\|^2. \quad (21)$$

Observing that both x_{t+1} and x_T^* are points in $\mathcal{K}_t$ (which implies $\delta_{\mathcal{K}_t}(x_{t+1}) = \delta_{\mathcal{K}_t}(x_T^*) = 0$), we can re-arrange (21) as

$$\begin{aligned} \frac{1}{2\eta_t} \|x_{t+1} - x_T^*\|^2 &\leq \frac{1}{2\eta_t} \|x_t - x_T^*\|^2 + \langle \nabla f_t(x_t), x_T^* - x_t \rangle \\ &\quad - \frac{1}{2\eta_t} \|x_t - x_{t+1}\|^2 + \langle \nabla f_t(x_t), x_t - x_{t+1} \rangle. \end{aligned} \quad (22)$$

We now use the easily-checked fact

$$\max \left\{ -\frac{1}{2\eta} \|u\|^2 + \langle v, u \rangle : u \in \mathbb{R}^n \right\} = \frac{\eta}{2} \|v\|^2,$$

which holds for all $\eta > 0$ and $v \in \mathbb{R}^n$, and amounts simply to compute the peak value of the concave quadratic function inside the braces. Using this fact with $\eta = \eta_t$ and $v = \nabla f_t(x_t)$, we can bound the sum of the last two terms on the right-hand side of (22), thereby arriving at

$$\frac{1}{2\eta_t} \|x_{t+1} - x_T^*\|^2 \leq \frac{1}{2\eta_t} \|x_t - x_T^*\|^2 + \frac{\eta_t}{2} \|\nabla f_t(x_t)\|^2 + \langle \nabla f_t(x_t), x_T^* - x_t \rangle. \quad (23)$$

The convexity of f_t implies $\langle \nabla f_t(x_t), x_T^* - x_t \rangle \leq f(x_T^*) - f(x_t)$ (e.g., see (7) with $m = 0$), which, when plugged in (23), yields

$$f_t(x_t) - f_t(x_T^*) \leq \frac{1}{2\eta_t} \|x_t - x_T^*\|^2 - \frac{1}{2\eta_t} \|x_{t+1} - x_T^*\|^2 + \frac{\eta_t}{2} \|\nabla f_t(x_t)\|^2. \quad (24)$$

For $t = 1$, inequality (24) implies

$$f_1(x_1) - f_1(x_T^*) \leq \frac{1}{2\eta_1} D^2 - \frac{1}{2\eta_1} \|x_2 - x_T^*\|^2 + \frac{\eta_1}{2} \|\nabla f_1(x_1)\|^2, \quad (25)$$

where Assumption 3.4 was used to bound the first term in the right-hand side of (24). For $t \geq 2$, we have

$$\begin{aligned} \frac{1}{2\eta_t} \|x_t - x_T^*\|^2 &= \frac{1}{2\eta_{t-1}} \|x_t - x_T^*\|^2 + \left(\frac{1}{2\eta_t} - \frac{1}{2\eta_{t-1}} \right) \|x_t - x_T^*\|^2 \\ &\leq \frac{1}{2\eta_{t-1}} \|x_t - x_T^*\|^2 + \left(\frac{1}{2\eta_t} - \frac{1}{2\eta_{t-1}} \right) D^2, \end{aligned} \quad (26)$$

where the last inequality uses Assumptions 3.4 and 3.7. Plugging (26) in (24) gives

$$\begin{aligned} f_t(x_t) - f_t(x_T^*) &\leq \frac{1}{2\eta_{t-1}} \|x_t - x_T^*\|^2 - \frac{1}{2\eta_t} \|x_{t+1} - x_T^*\|^2 \\ &\quad + \frac{\eta_t}{2} \|\nabla f_t(x_t)\|^2 + \left(\frac{1}{2\eta_t} - \frac{1}{2\eta_{t-1}} \right) D^2, \end{aligned} \quad (27)$$

which is valid for $t \geq 2$. Finally, summing (25) to (27) (instantiated from $t = 2$ to T) yields

$$\begin{aligned} \text{Regret}_T &\leq \frac{1}{2\eta_T} D^2 + \sum_{t=1}^T \frac{\eta_t}{2} \|\nabla f_t(x_t)\|^2 \\ &\leq \frac{T^\beta D^2}{2} + \frac{L^2}{2} \sum_{t=1}^T \frac{1}{t^\beta} \\ &\leq O(T^{\max\{\beta, 1-\beta\}}), \end{aligned}$$

where the second inequality follows from Assumption 3.5 and the choice of step-size sequence.

We now turn to the analysis of $\text{CCV}_{T,2}$: combining Lemma 4.3 with the choice of step-size sequence gives directly $\text{CCV}_{T,2} \leq O(T^{1-\beta})$ and concludes the proof.

C. Proof of Theorem 4.5

Looking first at Regret_T and replaying the opening moves in the proof of Theorem 4.4, we may assume that inequality (23) holds, an inequality that we reproduce here for the convenience of the reader:

$$\frac{1}{2\eta_t} \|x_{t+1} - x_T^*\|^2 \leq \frac{1}{2\eta_t} \|x_t - x_T^*\|^2 + \frac{\eta_t}{2} \|\nabla f_t(x_t)\|^2 + \langle \nabla f_t(x_t), x_T^* - x_t \rangle. \quad (28)$$

The strong convexity of f_t implies, via (8),

$$\langle f_t(x_t), x_T^* - x_t \rangle \leq f_t(x_T^*) - f_t(x_t) - (m/2) \|x_t - x_T^*\|^2,$$

which, when inserted in (28), gives

$$f_t(x_t) - f_t(x_T^*) \leq \frac{m(t-1)}{2} \|x_t - x_T^*\|^2 - \frac{mt}{2} \|x_{t+1} - x_T^*\|^2 + \frac{\eta_t}{2} \|\nabla f_t(x_t)\|^2 \quad (29)$$

(where we used the choice of step-size sequence to replace η_t in some terms). Adding (29) from $t = 1$ to T yields

$$\begin{aligned} \text{Regret}_T &\leq \sum_{t=1}^T \frac{\eta_t}{2} \|\nabla f_t(x_t)\|^2 \\ &\leq \frac{L^2}{2} \sum_{t=1}^T \frac{1}{mt} \\ &= O(\log T), \end{aligned}$$

where the second inequality is due to Assumption 3.5 and the choice of step-size sequence. It remains to account for $\text{CCV}_{T,2}$. For this, concatenate Lemma 4.3 with the choice of step-size sequence to get $\text{CCV}_{T,2} \leq O(\log T)$, thereby closing the proof.

D. Proof of Theorem 5.3

The bound on $\text{CCV}_{T,2}$ is straightforward from Lemma 5.1, by using the chosen step-size $1/t^\beta$, with $\beta \in (0, 1)$.

Regarding the regret bound, we assume that the loss functions f_t are convex, thus the modulus of convexity m is zero, as explained in Section 3.2. Then, from Lemma 5.2, we can rewrite the inequality as

$$\text{Regret}_T \leq \sum_{t=1}^T \frac{1}{2\eta_t} \left(\|x_t - x_T^*\|^2 - \|x_{t+1} - x_T^*\|^2 \right) + \frac{3}{2} L^2 \sum_{t=1}^T \eta_t. \quad (30)$$

Note that

$$\begin{aligned} &\sum_{t=1}^T \frac{1}{2\eta_t} \left(\|x_t - x_T^*\|^2 - \|x_{t+1} - x_T^*\|^2 \right) \\ &\leq \frac{1}{2\eta_1} \|x_1 - x_T^*\|^2 + \sum_{t=2}^T \left(\frac{1}{2\eta_t} - \frac{1}{2\eta_{t-1}} \right) \|x_t - x_T^*\|^2 \\ &\leq \frac{1}{2\eta_T} D^2 \end{aligned}$$

where the first inequality results from Assumption 3.7 and the nonnegativity of the norm; and the second inequality derives from the combination of Assumptions 3.4 and 3.7, from which we know that $\forall t \geq 1$, $\|x_t - x_T^*\| \leq D$ and $\left(\frac{1}{2\eta_t} - \frac{1}{2\eta_{t-1}} \right) \geq 0$. Plugging this result into (30), we obtain

$$\begin{aligned} \text{Regret}_T &\leq \frac{1}{2\eta_T} D^2 + \frac{3}{2} L^2 \sum_{t=1}^T \eta_t \\ &\leq O(T^{\max\{\beta, 1-\beta\}}). \end{aligned} \quad (31)$$

where the last inequality is due to the choice of the step-size sequence.

E. Proof of Theorem 5.4

The bound on $\text{CCV}_{T,2}$ is straightforward from Lemma 5.1, by using the chosen step-size $\eta_t = 1/(mt)$.

Regarding the regret bound, in this case, we assume that $m > 0$. Similarly to the proof of Theorem 5.3, notice that we can rewrite the regret bound of Lemma 5.2 as

$$\text{Regret}_T \leq \left(\frac{1}{2\eta_1} - \frac{m}{2} \right) \|x_1 - x_T^*\|^2 + \sum_{t=2}^T \left(\frac{1}{2\eta_t} - \frac{1}{2\eta_{t-1}} - \frac{m}{2} \right) \|x_t - x_T^*\|^2 + \frac{3}{2} L^2 \sum_{t=1}^T \eta_t,$$

due to the nonnegativity of the norm. By the choice of step-size, we have

$$\text{Regret}_T \leq \frac{3}{2} L^2 \sum_{t=1}^T \frac{1}{mt} \leq O(\log T).$$

E.1. Additional Numerical Experiments

In this section, we present additional numerical experiments. In particular, a version of the Online Linear Regression experiment with additional rounds, and an experiment for Online Support Vector Machine. The results were obtained by averaging over 50 trials and reported with a 95% confidence interval.

E.1.1. ONLINE LINEAR REGRESSION WITH ADDITIONAL ROUNDS

We now present a version of the Online Linear Regression experiment with a larger number of rounds. The Switch algorithm is memory-intensive as the set to which the algorithm performs the projections incorporates all previously revealed constraint functions. Therefore, as the iterations unfold, the set becomes increasingly more complex and the associated projection increasingly more expensive. So, we removed Switch from this experiment to be able to investigate a larger number of rounds.

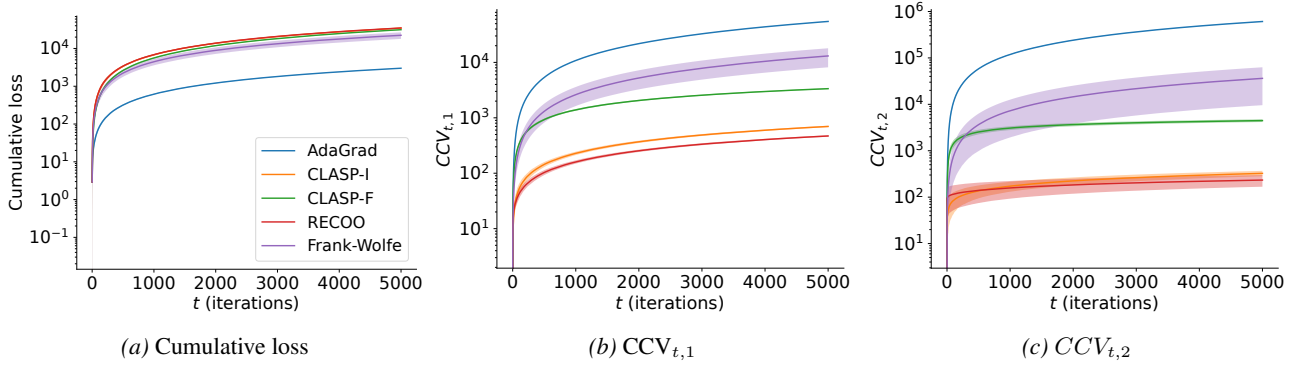


Figure 2. Online Linear Regression with additional rounds.

The results are similar to the ones in Section 6. However, while the Cumulative Constraint Violation (both $\text{CCV}_{T,1}$ and $\text{CCV}_{T,2}$) attained by CLASP and RECOO are very similar, note that, for the $\text{CCV}_{T,2}$ metric, for a larger number of rounds, the CLASP algorithm displays diminished variability. Furthermore, note that for a larger number of rounds, we see that CLASP-F attains lower Cumulative Constraint Violation (both $\text{CCV}_{T,1}$ and $\text{CCV}_{T,2}$), something that cannot be inferred from the previous experiment. These results demonstrate the capability of our proposed approach to keep violation severity sublinear while maintaining reasonable regret.

E.1.2. ONLINE SUPPORT VECTOR MACHINE

In our second experiment, we compare the performance of the algorithms for the online update of a Support Vector Machine (SVM) (Boser et al., 1992; Bishop, 2006). We consider that, at each iteration t , we receive a new labeled sample (u_t, v_t) , with $u_t \in \mathbb{R}^P$ the feature vector and $v_t \in \{\pm 1\}$ the label. Thus, at each round t , we can formulate the optimization problem in the COCO setting as

$$\underset{x:=(w,b) \in \mathcal{K}}{\text{minimize}} \quad \frac{1}{2} \|w\|^2, \quad \text{subject to} \quad -v_t (w^T u_t - b) + 1 \leq 0,$$

or, in terms of the COCO framework, the revealed loss function is $f_t(x) = \frac{1}{2} \|w\|^2$ and the revealed constraint function is $g_t(x) = -v_t (w^T u_t - b) + 1$. For this experiment, we use the real-world dataset about wine quality from their physicochemical properties (for the details about the dataset, see Cortez et al. (2009))². The dataset contains 6497 samples, each sample contains $P = 11$ features, and the quality of the wine is classified between 1 and 9. Some of the features were

²This dataset is released under a Creative Commons Attribution 4.0 International (CC BY 4.0) license.

in g/dm^3 , while others were in mg/dm^3 . We ensured that all density features were expressed in g/dm^3 . In this experiment, we consider the binary classification setting, where we label with 1 the wine samples with quality equal to or greater than 7, and label with -1 the remaining. Thus, we want our classifier to distinguish between high-quality and low-quality wines. In each trial, we reshuffle the dataset so that the order of samples in each trial is always different. The vector $x_1 \in \mathbb{R}^{P+1}$ for all algorithms was initialized such that $x_i \sim U(-1, 1)$, for $1 \leq i \leq P+1$.

From domain knowledge, we can bound the norm of the feature vector as each physicochemical property has an interval of possible values (for simplicity, we analyzed the possible values encountered in the dataset and concluded that $0 \leq u_t \leq 70$ for all t , thus we can use this to define the Lipschitz constant of the constraint functions $L = 70\sqrt{P}$). While the SVM is an unconstrained problem, our algorithms assume that the decision set is compact; thus, based on the constraints on the feature vectors, we define $\mathcal{K} = \{x = (w, b) \in \mathbb{R}^{P+1} : \|x\| \leq 70\sqrt{P}\}$.

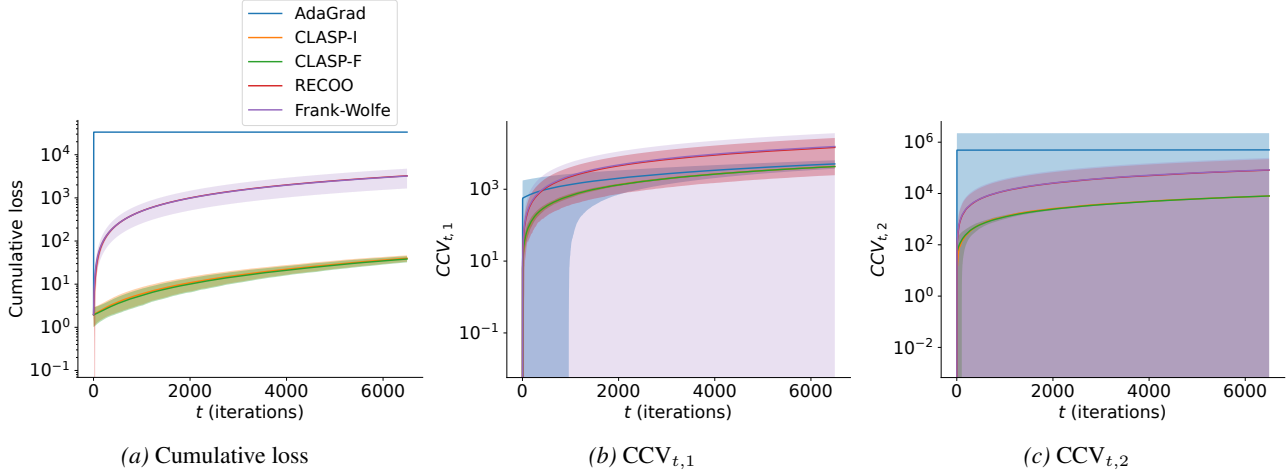


Figure 3. Online Support Vector Machine

Remark. In this experiment, the feasibility assumption is not satisfied, i.e., the set $\mathcal{K} \cap \mathcal{C}_T = \emptyset$, where $\mathcal{C}_T = \bigcap_{t=1}^T \mathcal{C}_t$. Although this property is used in the analysis of the algorithms in COCO to obtain regret and cumulative constraint violation bounds, most of the algorithms in COCO can still be applied to problems without this property. However, the Switch algorithm, due to the exploitation of nested convex bodies, cannot be applied when the feasibility property is not satisfied. Since, at each iteration, the algorithm projects onto the intersection of past constraint functions, then there will be some T_0 , with $1 \leq T_0 \leq T$, such that $\mathcal{C}_{T_0} = \bigcap_{t=1}^{T_0} \mathcal{C}_t = \emptyset$. Therefore, in this experiment, we cannot compare the performance of the Switch algorithm, but compare our algorithms, CLASP-I and CLASP-F, with the AdaGrad, RECOO and Frank-Wolfe algorithms.

In Fig. 3, we can visualize both the cumulative loss and the CCV of the different algorithms for each round. In this experiment, the more informative results are in Figs. 3b and 3c, as a constraint violation translates as a transgression of the margin. We see for both constraint violation metrics, $CCV_{T,1}$ and $CCV_{T,2}$, our proposed approach, CLASP-I and CLASP-F, achieves better results and with less variability. Furthermore, we see that for the metric $CCV_{T,2}$, the difference in performance is significantly better.

E.1.3. DRONE FLIGHT WITH ADVERSARIAL CHANGES

In this section, we formulate a drone flight problem with adversarial constraints. Consider a drone and, at each time step t , we want the drone to maintain a specific velocity vector $v_t \in \mathbb{R}^d$, which can change in direction over time. Over time, conditions can change, such as the winds. Therefore, at each time step, we can consider that safety bounds on the speed of the drone in each axis are revealed, i.e., two vectors $v_t^{\text{lower}}, v_t^{\text{upper}} \in \mathbb{R}^d$ such that $v_t^{\text{lower}} \leq v_t \leq v_t^{\text{upper}}$. Furthermore, we can consider a static decision set $\mathcal{K} := \{v \in \mathbb{R}^d : \|v\|^2 \leq c\}$, which is associated with the drone characteristic and bounds the maximum speed the drone can achieve. In this experiment, we consider $c = 4$. Thus, at each round t , we can formulate the

optimization problem in the COCO setting as

$$\underset{v \in \mathcal{K}}{\text{minimize}} \quad \frac{1}{2} \|v - v_t\|^2, \quad \text{subject to} \quad Av - v_t^{\text{safe}} \leq 0,$$

where $A = \begin{bmatrix} I \\ -I \end{bmatrix}$, $v_t^{\text{safe}} = \begin{bmatrix} v_t^{\text{upper}} \\ -v_t^{\text{lower}} \end{bmatrix}$, the revealed loss function is $f_t(x) = \frac{1}{2} \|v - v_t\|^2$ and the revealed constraint function is $g_t(x) = Av - v_t^{\text{safe}}$. In this experiment, we consider $v_{t,i}^{\text{upper}} \sim U(0.1, 3)$ and $v_{t,i}^{\text{lower}} \sim U(-3, -0.1)$, for $1 \leq i \leq d$. Furthermore, we consider a two-dimensional space, i.e., $d = 2$. The velocity vector we intend to maintain at each iteration, v_t , will be generated by rotating the vector $v_0 := (0, 2)$ by some angle $\theta_t \sim U(-\frac{\pi}{6}, \frac{\pi}{6})$.

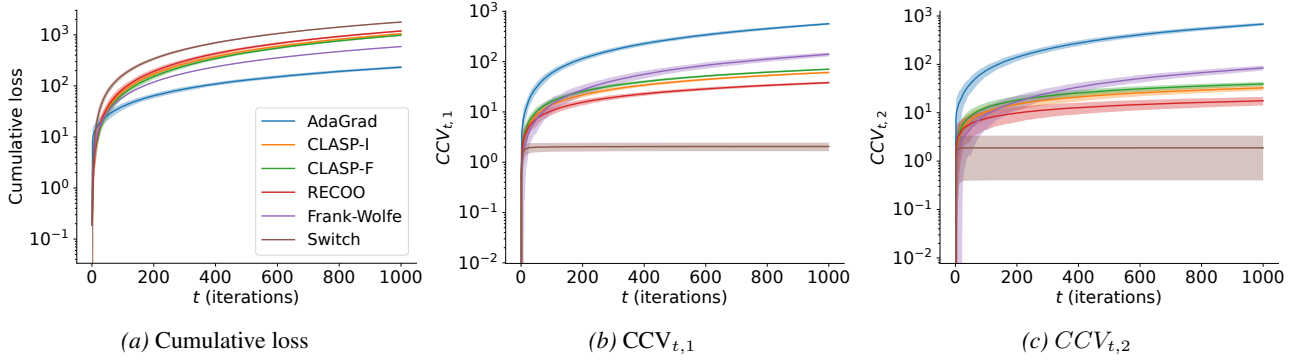


Figure 4. Drone flight with adversarial changes.

In Fig. 4, we summarize the results of the different algorithms. We see that the AdaGrad-based method achieves the lowest cumulative loss, at the cost of larger cumulative constraint violations (both $CCV_{T,1}$ and $CCV_{T,2}$). In contrast, Switch achieves the lowest cumulative constraint violations, at the cost of higher cumulative loss. Furthermore, we see that CLASP-I and CLASP-F present a good trade-off between cumulative loss and cumulative constraint violations, with results comparable to RECOO. The Frank-Wolfe-based algorithm achieves slightly lower cumulative loss than CLASP-I, CLASP-F, and RECOO, at the cost of higher cumulative constraint violation.